Algebra 1 | Name____Answer Key_____
Unit 2 | Date____________________
Lesson 5 Homework | Period___________________

1. Are the terms $-4xy^2$ and $\frac{1}{2}x^2y$ like terms? Why or why not? Explain.

No, exponents of each variable are different.

2. Find the sum of $(6 - t - t^2)$ and $(10t + t^2)$.

$(6 - t - t^2) + (10t + t^2) = 9t + 6$

3. Write the sum from question 2 in standard form.

$9t + 6$

4. Find the difference of $(5.5n^2 + 8 - 3n)$ and $(-3n^2 - 5)$.

$(5.5n^2 + 8 - 3n) - (-3n^2 - 5)$

$8.5n^2 + 13 - 3n$

5. Write the difference from question 4 in standard form.

$8.5n^2 - 3n + 13$

6. What polynomial must be added to $-5c^2 - 3 + 4c$ in order to get a sum of $-2c^2 - c + 1$?

$(-2\mathrm{c}^2 - \mathrm{c} + 1) - (-5\mathrm{c}^2 - 3 + 4\mathrm{c})$

$3c^2 - 5c + 4$

For questions 7-10, find each sum or difference. Write the result in standard form.

7. $(-3.4g^2h - 5gh^2 + 7) + (-1.75gh^2 - 12)$

$-3.4g^2h - 6.75gh^2 - 5$

8. $2\frac{1}{5}r^5t^2 - (3r^4t^3 + r^5t^2 - 6rt^4)$

$\frac{11}{5}r^5t^2 - 3r^4t^3 - r^5t^2 + 6rt^4$

$\frac{6}{5}r^5t^2 - 3r^4t^3 + 6rt^4$

9. $(8a^2 - 3a + 11b^2) + (2a - 5b^2 - 2a^2)$

$6a^2 + 6b^2 - a$

10. $(9.3m^2 - 5.1n) - (2n^2 + 6.75m)$

$9.3m^2 - 2n^2 - 6.75m - 5.1n$